Chapter II: A Model of Production

Jesús Bueren

Católica-Lisbon

Two Ways to Answer Big Economic Questions

- There are two main approaches in economics:
- **1. Empirical approach**
  - Use data to estimate causal effects.
  - Look for natural experiments.
  - Compare treated vs. control groups.
- **2. Theoretical approach**
  - Build a model of the whole economy.
  - Make assumptions explicit.
  - Study how all markets adjust simultaneously.
- If possible, combining both approaches is the best way to answer questions in economics.

Why Isn't Empirical Evidence Enough?

- First problem: **No natural experiment for everything**
  - We cannot randomly assign an economy to "AI revolution".
- Second problem: equilibrium effects
  - A policy affects one group...
  - But prices adjust for everyone
  - Other markets respond.
  - No control group
- To understand how prices and quantities adjust, we need a model.

An Economic Model Is Like a Video Game

- A model is just a mathematical representation of a hypothetical world.
- Imagine designing a simple video game:
  - You create a world and give it resources.
  - You specify what the characters can do and how they make decisions.
  - You work out what happens when they interact.
- Building an economic model follows the same logic:
  - Specify the available workers, machines, and technology.
  - Describe how households and firms make decisions.
  - Find the resulting output, wages, and rental rates.
- We can then use the video game to answer questions about the world we created.
- In this chapter, we will build a simple model of production and use it to understand why some countries are rich and others are poor.

Production Function

- First step: describe how the economy produces output.
- Economists summarize this with a production function:

$$Y = F(K, L)$$

- Output depends on:
  - L — workers (labor)
  - K — machines
- This simple relationship will not only determine output determine wages and profits.

What Happens When We Add More Workers?

- Hold capital fixed (K constant).
- Suppose the labor force increases (migration) or decreases (aging).
- We assume that adding more workers has two effects:
  - Total output rises — more hands help produce more.
  - But each additional worker adds a bit less than the previous one — they share the same capital and tools.
- Mathematically, we capture this with:
  - Positive marginal product: $\frac{\partial F}{\partial L} > 0$
  - Diminishing marginal product: $\frac{\partial^2 F}{\partial L^2} < 0$

What Happens When We Add More Machines?

- Hold labor fixed (L constant).
- Suppose the firm invests in more machines or equipment.
- We assume that adding more capital has two effects:
  - Total output rises — more machines help produce more.
  - But each additional machine adds a bit less than the previous one — workers have only so much to operate.
- Mathematically, we capture this with:
  - Positive marginal product: $\frac{\partial F}{\partial K} > 0$
  - Diminishing marginal product: $\frac{\partial^2 F}{\partial K^2} < 0$

Cobb–Douglas Production Function

- A very popular production function in macro is the Cobb–Douglas:

$$Y = AK^{\alpha}L^{1-\alpha}$$

- A measures productivity: how good the economy is at turning inputs into output.
- The marginal products are easy to compute:

$$\frac{\partial Y}{\partial L} = (1-\alpha)AK^{\alpha}L^{-\alpha} \geq 0, \qquad \frac{\partial Y}{\partial K} = \alpha AK^{\alpha-1}L^{1-\alpha} \geq 0$$

- And both marginal products get smaller as we increase that input:

$$\frac{\partial^2 Y}{\partial L^2} \leq 0, \qquad \frac{\partial^2 Y}{\partial K^2} \leq 0$$

- The two inputs also help each other:

$$\frac{\partial^2 Y}{\partial K \partial L} = \alpha(1-\alpha)AK^{\alpha-1}L^{-\alpha} \geq 0$$

Adding machinery raises how much extra output you get from hiring an extra worker

Returns to Scale: What Happens if We Grow Everything?

- Recall our Cobb–Douglas technology:

$$Y = AK^{\alpha}L^{1-\alpha}$$

- What happens if we double both capital and labor?

$$F(2K, 2L) = A(2K)^{\alpha}(2L)^{1-\alpha} = 2\,F(K, L)$$

- Doubling inputs gives exactly double output — this is called **constant returns to scale**.

Why Constant Returns to Scale?

- For Cobb–Douglas, returns to scale depend on the sum of the exponents:

$$\alpha + (1 - \alpha) = 1$$

- If the sum is:
  - Equal to 1: constant returns to scale
  - Greater than 1: increasing returns
  - Less than 1: decreasing returns

Firm's Problem

- Second step: describe how firms choose labor and capital.
- So far, we have seen how output responds to changes in labor and capital.
- Firms choose K and L to maximize profits:

$$\max_{K,L} \; F(K,L) - rK - wL$$

- Profits = production minus cost of inputs.
- We set the price of output to 1 (numeraire).
- r = rental price of capital, w = wage of workers.
- Firms take r and w as given because markets are competitive (simplifying assumption — left-wing critique possible).

Firm's Problem: Solution

- Profit-maximization problem:

$$\max_{K,L} \Pi(K, L) = \max_{K,L} \; F(K, L) - rK - wL$$

  - Step 1: Choose optimal capital K (hold L fixed)
  - Step 2: Choose optimal labor L (hold K fixed)

- We will see that the solution satisfies:

$$\text{MP}_K(K^*, L^*) = \frac{\partial F(K^*, L^*)}{\partial K} = r, \quad \text{MP}_L(K^*, L^*) = \frac{\partial F(K^*, L^*)}{\partial L} = w$$

  Just like in micro: hire/rent until marginal revenue = marginal cost.

- MP_K and MP_L are the marginal product of capital and labor.

Maximization in Two Variables

- General math rule:

$$\max_{x,y} f(x, y)$$

- Solution satisfies first-order conditions:

$$\frac{\partial f(x^*, y^*)}{\partial x} = 0, \quad \frac{\partial f(x^*, y^*)}{\partial y} = 0$$

- Applied to the firm:

$$\frac{\partial \Pi(K^*, L^*)}{\partial K} = 0, \quad \frac{\partial \Pi(K^*, L^*)}{\partial L} = 0$$

- Hire each input until marginal revenue of the input = cost of the input (just like MR = MC in micro for one good).

Firm's Problem with Cobb–Douglas

- Profit function:

$$\Pi(K, L) = F(K, L) - rK - wL$$

- Cobb–Douglas:

$$\Pi(K, L) = AK^{\alpha}L^{1-\alpha} - rK - wL$$

- Goal: choose K and L to maximize profits.
- Taking the derivative and equating to zero we get:

$$\alpha AK^{\alpha-1}L^{1-\alpha} - r = 0, \quad (1-\alpha)AK^{\alpha}L^{-\alpha} - w = 0$$

$$\mathrm{MP}_K = \alpha AK^{\alpha-1}L^{1-\alpha} = r, \quad \mathrm{MP}_L = (1-\alpha)AK^{\alpha}L^{-\alpha} = w$$

- Hire inputs until MP = price (marginal revenue = marginal cost).

Firm's Problem

Example: Hiring Labor with Fixed Capital

- Imagine that you own 1 machine and decide how many workers to hire.
- Profit function:
$$\Pi(K, L) = AK^{\alpha}L^{1-\alpha} - wL$$
- FOC for labor:
$$\frac{\partial \Pi}{\partial L} = A(1-\alpha)K^{\alpha}L^{-\alpha} - w = 0$$
- Numbers: $A = 1, \alpha = 0.5, K = 1, w = 0.25$
$$0.5 \cdot 1^{0.5}L^{-0.5} - 0.25 = 0 \implies L = 4$$
- Hire workers until marginal product of labor = wage.

Firm's Problem

Example: Hiring Labor with Fixed Capital

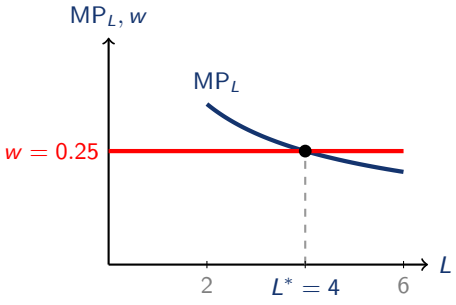
- Marginal product of labor:

$$\text{MP}_L = (1-\alpha)AK^{\alpha}L^{-\alpha}$$

- Downward sloping: each additional worker adds less output.
- Horizontal line = wage w.
- Hire labor until:

$$\text{MP}_L = w$$

- Intersection shows optimal number of workers L^*.
- This curve is called the labor demand curve.



Firm's Problem

Example: Renting Capital with Fixed Labor

- Imagine that you employ 1 worker and decide how many machines to rent.
- Profit function:

$$\Pi(K, L) = AK^{\alpha}L^{1-\alpha} - rK$$

- FOC for capital:

$$\frac{\partial \Pi}{\partial K} = \alpha A K^{\alpha-1} L^{1-\alpha} - r = 0$$

- Numbers: $A = 1, \alpha = 0.5, L = 1, r = 0.25$

$$0.5 \cdot 1^{0.5} K^{-0.5} - 0.25 = 0 \implies K = 4$$

- Rent machines until marginal product of capital = rental price.

Firm's Problem

Example: Renting Capital with Fixed Labor

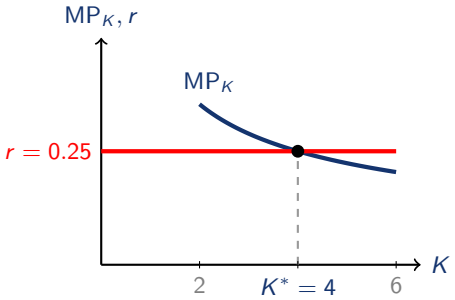
- Marginal product of capital:

$$\text{MP}_K = \alpha A K^{\alpha-1} L^{1-\alpha}$$

- Downward sloping: each additional machine adds less output.
- Horizontal line = rental price r.
- Rent capital until:

$$\text{MP}_K = r$$

- Intersection shows optimal number of machines K^*.
- This curve is called the capital demand curve.



Capital and Labor Supply

- We have seen how firms choose capital and labor to maximize profits
- Firms' decisions determine factor demand
- Now we introduce factor supply.
- Simplifying assumption:

$$K^s = \overline{K}, \qquad L^s = \overline{L}$$

(Fixed total amounts in the economy.)

Model Summary

- Last step: find the equilibrium.
- In **general equilibrium**, prices and quantities adjust so that all markets clear simultaneously. Our model has five equations:
- Capital demand: $\alpha A K^{\alpha-1} L^{1-\alpha} = r$
- Labor demand: $(1-\alpha) A K^{\alpha} L^{-\alpha} = w$
- Capital supply: $K = \overline{K}$
- Labor supply: $L = \overline{L}$
- Production function: $Y = A K^{\alpha} L^{1-\alpha}$
- The five endogenous variables are: K, L, r, w, and Y.

An Equilibrium

- In economics, the solution to a model is called an equilibrium.
- Equilibrium describes what happens when all markets clear, that is, supply equals demand in each market.
- In our model, we have five equations with five unknowns.
- Solving the model means finding the values of the endogenous variables (K, L, r, w, Y) in terms of the exogenous variables ($\overline{K}$, $\overline{L}$, A, α).
- This involves rewriting the system so that all endogenous variables are on one side of the equations and only parameters and exogenous variables on the other side.

Equilibrium Values

- In our simple model, the equilibrium values are:
- Capital: $K = \overline{K}$
- Labor: $L = \overline{L}$
- Rental rate of capital: $r = \alpha A \overline{K}^{\alpha-1} \overline{L}^{1-\alpha}$
- Wage: $w = (1-\alpha) A \overline{K}^{\alpha} \overline{L}^{-\alpha}$
- Output: $Y = A \overline{K}^{\alpha} \overline{L}^{1-\alpha}$
- These values satisfy all five equations of the model and clear labor, capital and goods markets.

Development Accounting

- We now make an important leap: we apply the production function of our model to aggregate economies across the world.
- We match output per person in the model (y) to GDP per capita in the data.
- Capital (k) is measured as the economy's stock of housing, factories, machines, and equipment, divided by population.
- The production function implies:

$$y = Ak^{1/3}.$$

- For now, we make a strong simplifying assumption:
  - Technology is the same across countries: A is constant across countries.

  so that income differences depend *only* on capital per person.

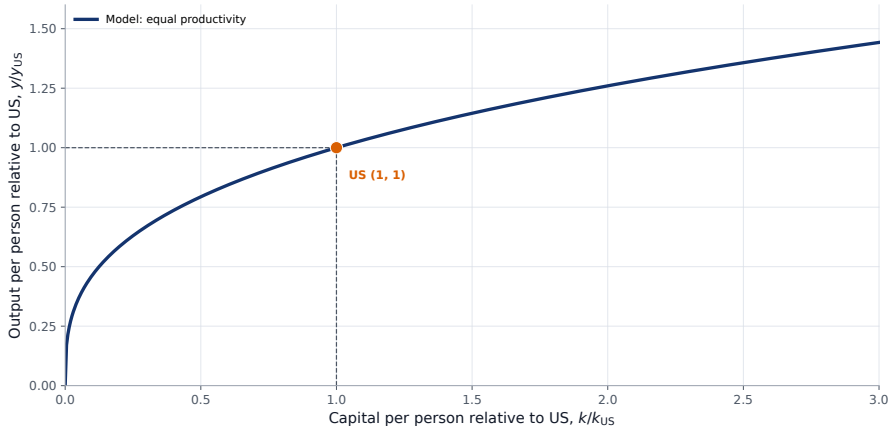
US as a benchmark

- We use the **United States as a benchmark** to compare countries.
- Measure each country's capital and output per person **relative to their US values**.
- With equal productivity across countries, the model implies:

$$\frac{y_i}{y_{\text{US}}} = \frac{Ak_i^{1/3}}{Ak_{\text{US}}^{1/3}} = \left(\frac{k_i}{k_{\text{US}}}\right)^{1/3}.$$

- A relative value of 1 means the same as the US; a value of 0.5 means half the US level.
- The US is therefore located at $(\mathbf{1}, \mathbf{1})$ on the graph.

The Model Relative to the US



Model: common productivity across countries. Capital and output per person are normalized relative to the US.

Why Is the Curve Increasing and Concave?

- The curve shows the model's prediction when productivity is held fixed:

$$\frac{y_i}{y_{\text{US}}} = \left(\frac{k_i}{k_{\text{US}}}\right)^{1/3}.$$

- **Increasing:** more capital per person allows more output per person.
- **Concave:** each additional unit of capital per person adds less output than the previous one.
- Changing capital per person gives:

$$\text{Doubling:} \quad 2^{1/3} \approx 1.26 \qquad \text{(about 26\% more output per person)},$$

$$\text{Halving:} \quad (1/2)^{1/3} \approx 0.79 \qquad \text{(about 21\% less output per person)}.$$

- This is **diminishing returns to capital**.

Predictions of the Model

Consider Portugal as an example.

- Capital per person in Portugal and Spain is close to or even slightly above the U.S. level.

According to the model:

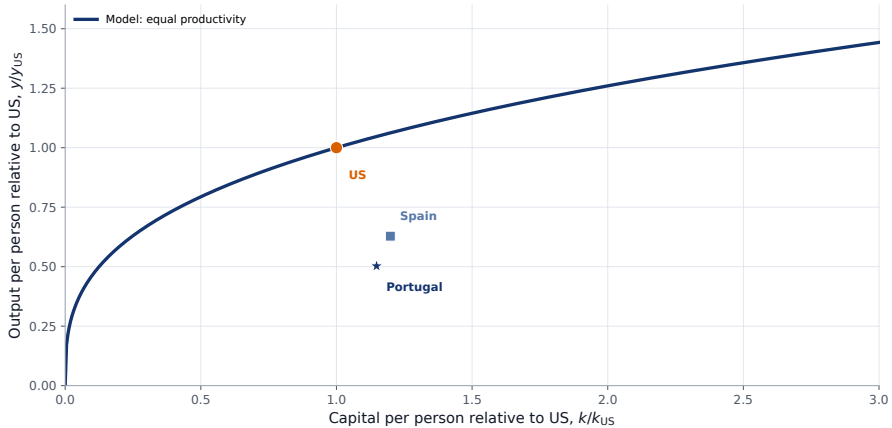
$$y = k^{1/3},$$

so if $k_{PRT} \approx k_{US}$, the model predicts:

$$y_{PRT} \approx y_{US}.$$

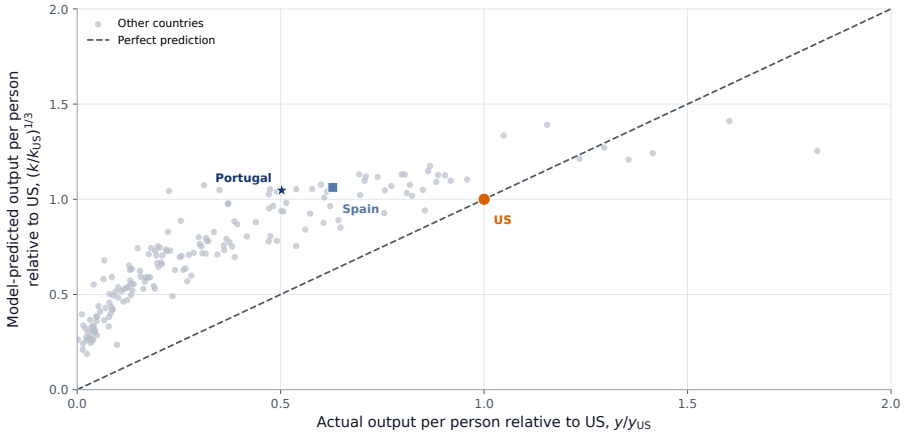
Therefore, according to the model, Portugal should be about as rich as the United States.

Production Model: Model vs Data



Source: Penn World Table 11.0 (Feenstra, Inklaar, and Timmer), 2023. Values are relative to the US.

Production Model: Model vs Data

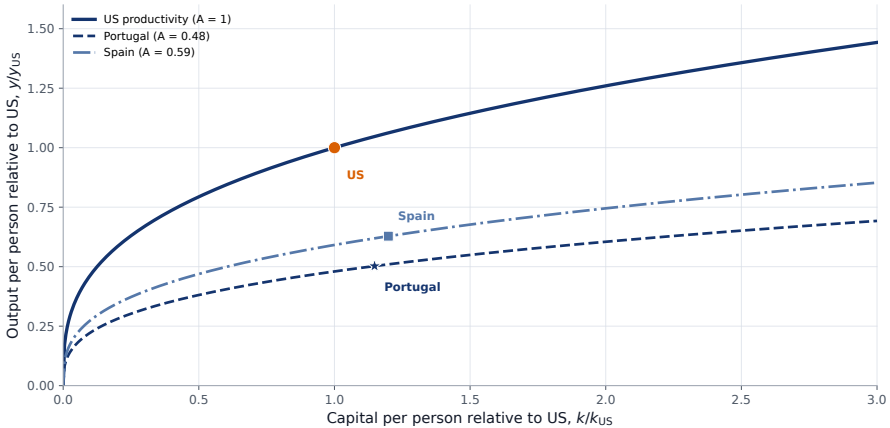


Source: Penn World Table 11.0 (Feenstra, Inklaar, and Timmer), 2023. Values are relative to the US.

Model vs Data

- In general, the model correctly predicts that countries are rich or poor according to how much capital per person they have.
- However, the magnitudes are very wrong: the model underestimates how poor countries are relative to the U.S.
- One way to reconcile the model with the data is to allow the productivity parameter A to be country-specific.
- A measures how productive countries are at using their factor inputs (in this case, K and L) to produce output.
- Thought experiment: what value of A_{PRT} would rationalize why Portugal produces much less output per capita than the U.S.?
- The inferred value of A is referred to as total factor productivity (TFP).

Production Model with Country-Specific TFP



Source: Penn World Table 11.0 (Feenstra, Inklaar, and Timmer), 2023. Values are relative to the US.

Poor vs Rich Countries

- Rich countries have about **125 times as much capital per person** as poor countries:

$$\frac{k_{\text{rich}}}{k_{\text{poor}}} \simeq 125.$$

- With equal productivity, this would predict only a **5-fold gap in output per person**: $125^{1/3} = 5$.

- In the data, output per person is about **70 times higher** in rich countries.

$$\underbrace{\frac{y_{\text{rich}}}{y_{\text{poor}}}}_{\simeq 70} = \underbrace{\frac{A_{\text{rich}}}{A_{\text{poor}}}}_{\simeq 14} \times \underbrace{\left(\frac{k_{\text{rich}}}{k_{\text{poor}}}\right)^{1/3}}_{\simeq 5}.$$

- The remaining gap is captured by **TFP** in the model: $A_{\text{rich}}/A_{\text{poor}} \simeq 70/5 = 14$.

Example: Cabo Verde

- The US has about **5 times as much capital per person** as Cabo Verde:

$$\frac{k_{US}}{k_{CPV}} \simeq 5.$$

- With equal productivity, $A_{US} = A_{CPV}$, the model predicts:

$$\frac{y_{US}}{y_{CPV}} = \frac{A_{US}}{A_{CPV}} \left(\frac{k_{US}}{k_{CPV}} \right)^{1/3} = 5^{1/3} \approx 1.71.$$

- In the data, the US has about **6.7 times as much output per person**:

$$\frac{y_{US}}{y_{CPV}} \simeq \frac{1}{0.15} \approx 6.7.$$

- Capital differences alone **underpredict the observed output gap**.

Example: Cabo Verde

- We infer **relative TFP** from the output and capital ratios:

$$\frac{A_{US}}{A_{CPV}} = \frac{y_{US}/y_{CPV}}{(k_{US}/k_{CPV})^{1/3}} \approx \frac{6.67}{1.71} \approx 3.9.$$

- Decomposing the income gap:

$$\underbrace{\frac{y_{US}}{y_{CPV}}}_{\simeq 6.7} = \underbrace{\frac{A_{US}}{A_{CPV}}}_{\simeq 3.9 \text{ (TFP)}} \times \underbrace{\left(\frac{k_{US}}{k_{CPV}}\right)^{1/3}}_{\simeq 1.7 \text{ (capital)}}.$$

- If Cabo Verde had US productivity, its output per person would be about **3.9 times higher**.
- If Cabo Verde had US capital per person, its output per person would be about **1.7 times higher**.
- The model attributes most of the output gap to **productivity differences**.

What Drives TFP Differences across Countries?

- In our simple model, A captures more than technology:
- **Education:** workers with more skills can produce more with the same physical capital. Our model captures this in A, since skills are not measured separately.
- **Institutions:** secure property rights and reliable rules encourage firms to invest, innovate, and expand.
- **Misallocation:** capital and labor may be stuck in less productive firms instead of moving to more productive ones.
- **Technology:** better production methods, and the ability to adopt them, raise output from the same inputs.